18 On Saturday, some adults and some children were in a theatre.
The ratio of the number of adults to the number of children was 5 : 2

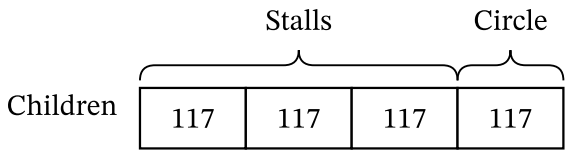
Each person had a seat in the Circle or had a seat in the Stalls.

$\frac{3}{4}$ of the children had seats in the Stalls.
117 children had seats in the Circle.

There are exactly 2600 seats in the theatre.

On this Saturday, were there people on more than 60% of the seats?
You must show how you get your answer.

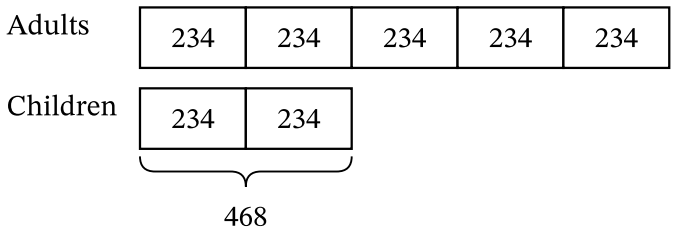
One quarter of the children were in the Circle.



1 mark
Find all the children

Total children = 117 × 4 = 468

2 ratio parts = 468, so 1 part = 234.

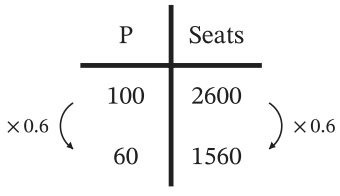


1 mark
Use adults : children as 5 : 2

1 mark
1170 adults or 1638 people

Adults = 5 × 234 = 1170; total = 1638

P = percentage; compare with 60% of 2600.



1 mark
Find 60% of all seats

Yes. 1638 people > 1560 seats (60%).

1 mark
Correct conclusion supported by these figures

(Total for Question 18 is 5 marks)

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